Customer Churn Analysis — Final Report

Prepared by Nirmal, Business Analyst · Advanced Telecommunications Solutions (ATS)

Executive Summary

This report summarises findings from the Customer Churn Analysis project for Advanced Telecommunications Solutions (ATS), covering customer segmentation (clustering), churn prediction (predictive modelling), and the business implications of both.

Across 6,741 customers analysed, the overall churn rate is **26.57%**. Customer segmentation identified one segment — short-tenure, high-monthly-charge customers — with a churn rate of **48.8%**, nearly double the overall average and representing **31.5%** of the customer base. A predictive model (Logistic Regression) was able to identify likely churners with 76.5% recall and 81.8% ROC-AUC, giving the business a practical tool for flagging at-risk customers before they leave.

Headline recommendation: retention efforts should prioritise newer, higher-paying customers on short tenures, as this group accounts for the largest share of churn risk in the customer base.

This report draws on the completed clustering analysis and predictive modelling work to date, and presents the project's conclusions, recommendations and business implications.

Project Overview & Methodology

Objective: Identify the key factors driving customer churn for ATS and recommend targeted retention strategies, using customer segmentation and predictive modelling.

Team & roles:

- Project Manager — Sita
- Data Engineer (preprocessing & feature engineering) — Nithin
- Data Analyst, Clustering Analysis — Sangeeta
- Data Analyst, Predictive Modelling — Shreeji
- Business Analyst — Nirmal (this report)

Approach:

1. **Data Preparation** — dataset cleaned, missing values handled, categorical variables encoded, numeric features scaled.

2. **Clustering Analysis** — K-Means clustering used to segment customers by tenure and monthly charges; optimal cluster count selected via the Elbow Method and Silhouette Score.
3. **Predictive Modelling** — Logistic Regression and Random Forest models trained to predict churn, evaluated on Accuracy, Precision, Recall, F1-score and ROC-AUC.
4. **Business Analysis** — translating clustering and modelling outputs into churn drivers, high-risk segments, and retention recommendations (this stage).

Customer Segmentation (Clustering) Findings

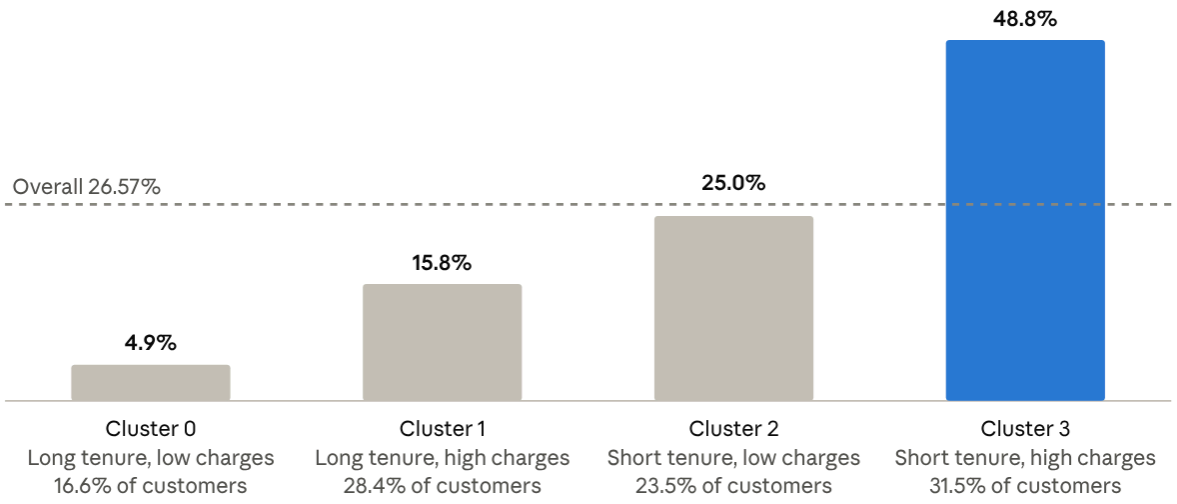
Sangeeta's clustering analysis used K-Means on **tenure** and **monthly charges** (the churn label itself was excluded from clustering inputs). Testing K=2 through K=10 via the Elbow Method and Silhouette Score, **K=4** was selected as the optimal number of clusters (silhouette score 0.47).

Cluster	Customers	% of Base	Avg Tenure (months)	Avg Monthly Charges	Churn Rate	Risk Level
0	1,118	16.6%	54.0	$34.54	4.9%	Low
1	1,914	28.4%	58.6	$93.03	15.8%	Medium
2	1,584	23.5%	11.2	$33.84	25.0%	Medium-High
3	2,125	31.5%	15.0	$81.68	48.8%	High

Overall churn rate across all customers: 26.57%.

Cluster 3 churns at 48.8%, nearly double the overall rate

Churn rate by customer segment · 6,741 customers



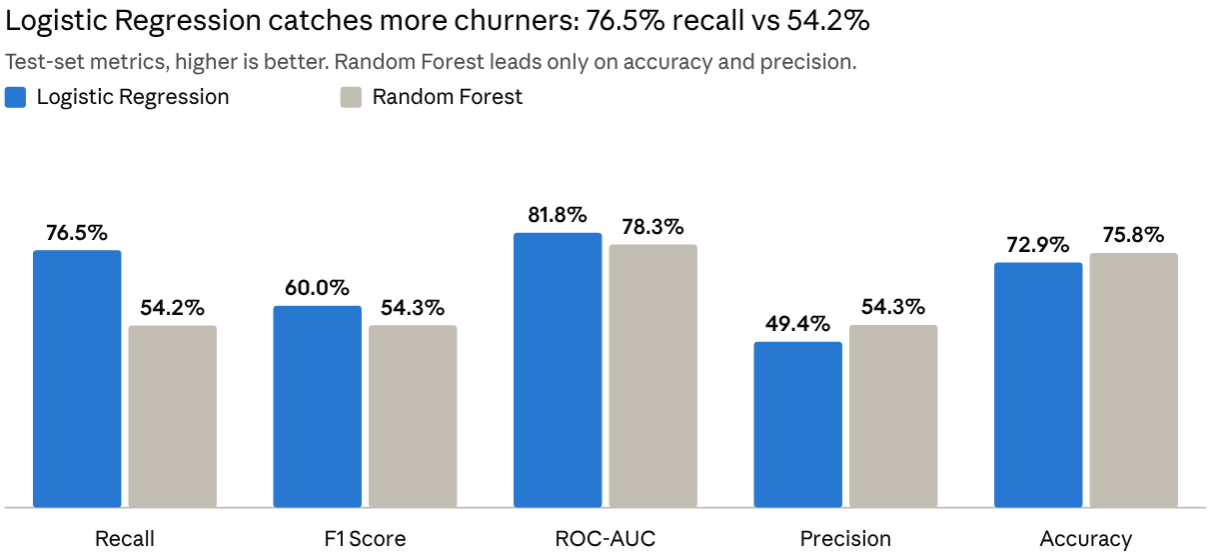
Source: Sangeeta's K-Means clustering output (Customer_Churn_Clustering_final.ipynb) · 4 segments, 6,741 customers; overall rate from the notebook

Scope note: this segmentation is based on tenure and monthly charges only. Cross-referencing these clusters against contract type and service usage would strengthen the business narrative and is recommended as a follow-up.

Predictive Modelling Findings

Shreeji trained and compared two classification models to predict customer churn:

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	72.94%	49.37%	**76.54%**	**60.02%**	**81.79%**
Random Forest	**75.76%**	**54.34%**	54.19%	54.27%	78.26%



Source: Shreeji's predictive modelling README (ATS_Predictive_Modelling) · test-set metrics for 2 models

Logistic Regression was selected as the preferred model, despite lower accuracy, because it achieved the highest Recall, F1 Score and ROC-AUC — meaning it is better at correctly catching customers who are actually going to churn, which matters more than raw accuracy for a retention use case (missing an actual churner is more costly than a false alarm).

The model generates a churn probability for each customer, grouped into **Low / Medium / High** risk bands, which can support targeted retention activity.

Note: feature importance / coefficient output from this model has not yet been generated from an executed run — see Limitations.

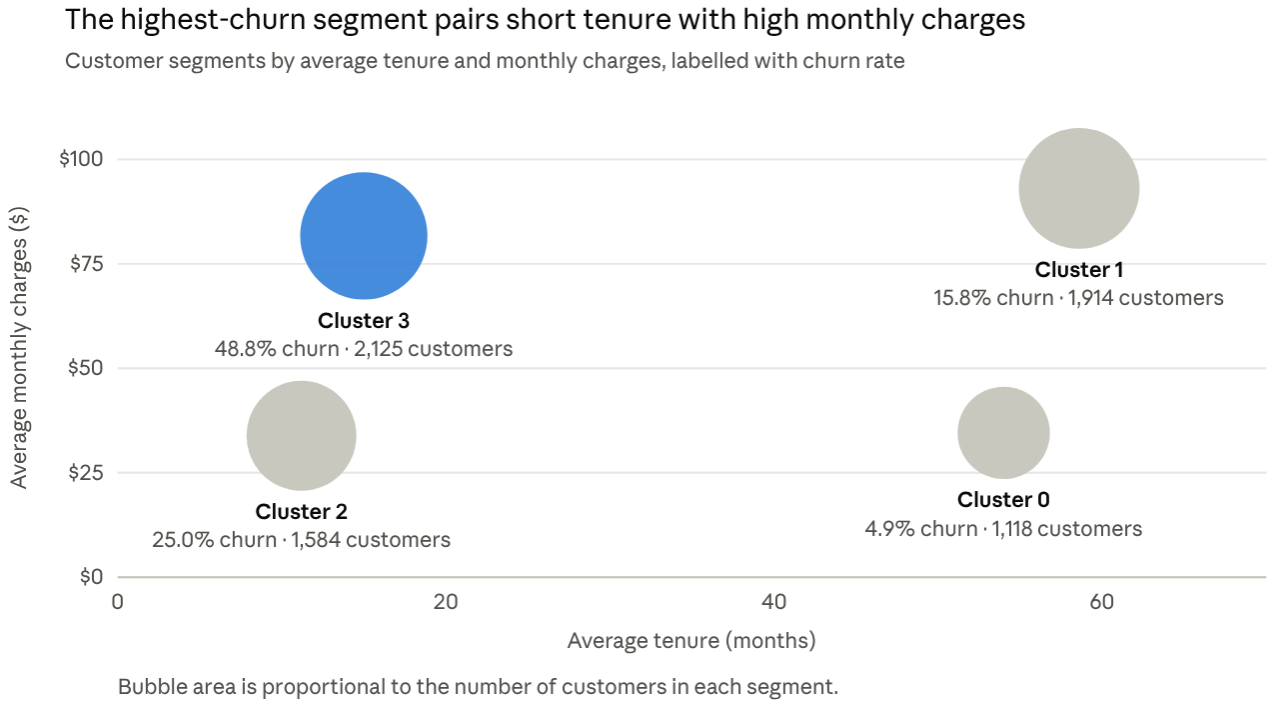
Key Churn Drivers

Based on findings available so far:

- **Tenure** — the single clearest driver. Customers with short tenure (~11-15 months) show substantially higher churn (25-48.8%) than long-tenure customers (~54-59 months, 4.9-15.8% churn).
- **Monthly charges** — higher-paying customers churn more when combined with short tenure (Cluster 3: $81.68 avg charge, 48.8% churn) than lower-paying, short-tenure customers (Cluster 2: $33.84 avg charge, 25.0% churn) — suggesting price sensitivity compounds the tenure effect.
- **Combined effect** — the highest-risk group is short-tenure *and* high-charge customers, not either factor alone.

To be completed once the predictive model's feature importance / coefficients (Logistic Regression) are generated from an executed run — this will confirm which additional fields (e.g. contract type, internet service, tech support) rank as predictive drivers, beyond tenure and charges.

High-Risk Customer Segments



Source: Sangeeta's K-Means clustering output (Customer_Churn_Clustering_final.ipynb) · cluster averages, 6,741 customers

Cluster 3 — Short tenure, high monthly charges is the priority segment:

- 2,125 customers (31.5% of the base)
- Average tenure: 15.0 months
- Average monthly charges: $81.68
- Churn rate: 48.8% — nearly double the overall rate of 26.57%

Cluster 2 — Short tenure, low monthly charges is a secondary watch segment:

- 1,584 customers (23.5% of the base)
- Churn rate: 25.0%

By contrast, **Cluster 0 — long tenure, low charges** is the safest segment (4.9% churn) and represents the retention benchmark the business should be working toward for other segments.

Recommended Retention Strategies

Based on the churn drivers and high-risk segments identified above, the following targeted strategies are recommended:

1. **Early-tenure retention program (Cluster 3 focus)** — introduce proactive check-ins, onboarding support and loyalty incentives for customers within their first 12-18 months, with particular focus on higher-paying customers who show the steepest churn risk.
2. **Value-perception review for high-charge, short-tenure customers** — review pricing, bundling and plan-fit for Cluster 3 customers specifically, since high monthly charges paired with short tenure is the strongest risk combination identified. Consider discounted bundles or step-up pricing that softens the early cost burden.
3. **Contract incentives** — encourage a shift from month-to-month arrangements toward longer-term contracts through sign-up incentives, since longer tenure is consistently associated with lower churn across all segments.
4. **Targeted monitoring for Cluster 2 (short tenure, lower charges)** — while lower risk than Cluster 3, this group still churns well above the long-tenure segments; lighter-touch engagement (e.g. satisfaction surveys, service usage nudges) is recommended here rather than high-cost intervention.
5. **Model-driven targeting** — use the Logistic Regression churn-probability scores (High / Medium / Low risk bands) to prioritise outreach operationally, rather than relying on segment averages alone.

Prioritised Recommendations

Ranked by expected impact versus effort to implement:

Priority	Recommendation	Expected Impact	Effort
1	Early-tenure retention program for Cluster 3	High — targets the largest, highest-risk segment (31.5% of customers, 48.8% churn)	Medium
2	Value-perception / pricing review for high-charge, short-tenure customers	High — directly addresses the strongest churn combination found	Medium-High
3	Model-driven targeting using churn-probability risk bands	High — low-cost, immediately actionable with existing model output	Low

Priority	Recommendation	Expected Impact	Effort
4	Contract incentives to extend tenure	Medium — long-term benefit, slower to show results	Medium
5	Lighter-touch monitoring for Cluster 2	Low-Medium — secondary segment, smaller risk gap	Low

Quick win: Priority 3 (model-driven targeting) can be actioned immediately using the existing Logistic Regression output and requires no new investment, making it the recommended starting point.

Limitations and Challenges

- **Predictive model not yet executed with real outputs.** The predictive modelling notebook is complete and correctly structured but has not been run end-to-end against the live dataset, so feature importance and individual churn-probability predictions are not yet available for this report. *Proposed solution:* request Shreeji run the notebook and share the executed outputs.
- **Clustering used only two features** (tenure, monthly charges). Other potentially relevant fields (contract type, internet service, tech support usage) were not included in the segmentation. *Proposed solution:* cross-tabulate the existing clusters against these fields for a richer segment profile.
- **Team availability.** Shreeji could not be reached during early team meetings, creating some uncertainty on the predictive modelling timeline. *Proposed solution:* establish a more consistent communication channel/check-in cadence.
- **Sequential dependency.** As Business Analyst, this report is dependent on both the Data Engineer and both Data Analysts delivering their outputs in sequence, which limits how far analysis can progress until each stage is complete.

Business Implications

Financial exposure: Cluster 3 (2,125 customers, 48.8% churn, average $81.68/month) represents the largest concentration of at-risk recurring revenue in the customer base. Even a modest reduction in this segment's churn rate would have an outsized effect on overall retention, given both its size and its above-average spend.

Expected benefit: Implementing the prioritised retention strategies — particularly early-tenure engagement and pricing/value review for Cluster 3 — is expected to reduce churn in the highest-risk segment and bring it closer to the behaviour seen in Cluster 1 (similar

spend level, but 58.6-month average tenure and only 15.8% churn), suggesting meaningful headroom for improvement.

Risks and trade-offs:

- Pricing or bundling changes carry margin trade-offs and should be modelled financially before rollout.
- Proactive outreach at scale requires operational capacity (staff time, contact channels) not yet costed in this report.
- Retention actions targeted using model risk-scores will include some false positives (customers flagged as high-risk who would not have churned), so outreach costs should be weighed against the model's precision (49.4%) alongside its recall.

Confidence caveat: These implications are based on correlational findings (clustering and predictive modelling), not controlled experiments. A/B testing any retention intervention before full rollout is recommended to confirm actual impact.

Next Steps

1. Get the predictive modelling notebook executed with real data to obtain feature importance / coefficients.
2. Cross-tabulate clustering segments against contract type and service usage fields.
3. Finalise Retention Strategies based on confirmed drivers.
4. Complete Prioritised Recommendations and Business Implications sections.
5. Prepare stakeholder presentation summarising final findings.